

Question 1

X_1 (Working hour)	X_2 (Sleep hours)	Y (Performance)
1	6	60
2	7	65
3	8	70

X_1 (Performance)	X_2 Working hours	X_3 Sleep hours
60	1	6
65	2	7
70	3	8

X_1	X_2	X_3	$X_1 X_2$	$X_1 X_3$	$X_2 X_3$	$(X_2)^2$	$(X_3)^2$	
60	1	6	60	360	6	1	36	
65	2	7	130	455	14	4	49	
70	3	8	210	560	24	9	64	
$\sum X_1 = 195$		$\sum X_2 = 6$	$\sum X_3 = 21$	$\sum X_1 X_2 = 400$	$\sum X_1 X_3 = 1375$	$\sum X_2 X_3 = 44$	$\sum (X_2)^2 = 14$	$\sum (X_3)^2 = 149$

$$\begin{aligned}\sum X_1 &= N \cdot a_{1.23} + b_{12.3} \cdot \sum X_2 + b_{13.2} \cdot \sum X_3 \\ \sum X_1 X_2 &= a_{1.23} \cdot \sum X_2 + b_{12.3} \cdot \sum (X_2)^2 + b_{13.2} \cdot \sum X_2 X_3 \\ \sum X_1 X_3 &= a_{1.23} \cdot \sum X_3 + b_{12.3} \cdot \sum X_2 X_3 + b_{13.2} \cdot \sum (X_3)^2\end{aligned}$$

$$195 = 3a_{1.23} + 6b_{12.3} + 21b_{13.2}$$

$$400 = 6a_{1.23} + 14b_{12.3} + 44b_{13.2}$$

$$1375 = 21a_{1.23} + 44b_{12.3} + 149b_{13.2}$$

(Infinite Solutions)

Question 2

A 44
B 57
C 62
D 49
E 71
F 53
G 60
H 58
I 65
J 73
K 55
L 92

Second std

$$\mu \pm 2s + d$$

$$[61.58 - 2(12.16), 61.58 + 2(12.16)]$$

$$[61.58 - 24.32, 61.58 + 24.32]$$

$$[37.26, 85.9] \leftarrow \text{Range}$$

Player L is outlier

$$\text{Mean} = \frac{44 + 57 + 62 + 49 + 71 + 53 + 60 + 58 + 65 + 73 + 55 + 92}{12}$$

$$12$$

$$= 61.58$$

Std

44 - 61.58	$(-17.58)^2$	=	309.05
57 - 61.58	$(-4.58)^2$	=	20.97
62 - 61.58	$(0.42)^2$	=	0.17
49 - 61.58	$(-12.58)^2$	=	158.25
71 - 61.58	$(9.42)^2$	=	88.73
53 - 61.58	$(-8.58)^2$	=	73.61
60 - 61.58	$(-1.58)^2$	=	2.49
58 - 61.58	$(-3.58)^2$	=	12.81
65 - 61.58	$(3.42)^2$	=	11.69
73 - 61.58	$(11.42)^2$	=	130.41
55 - 61.58	$(-6.58)^2$	=	43.29
92 - 61.58	$(30.42)^2$	=	925.37

$$\sum (x - \mu)^2 = 1776.84$$

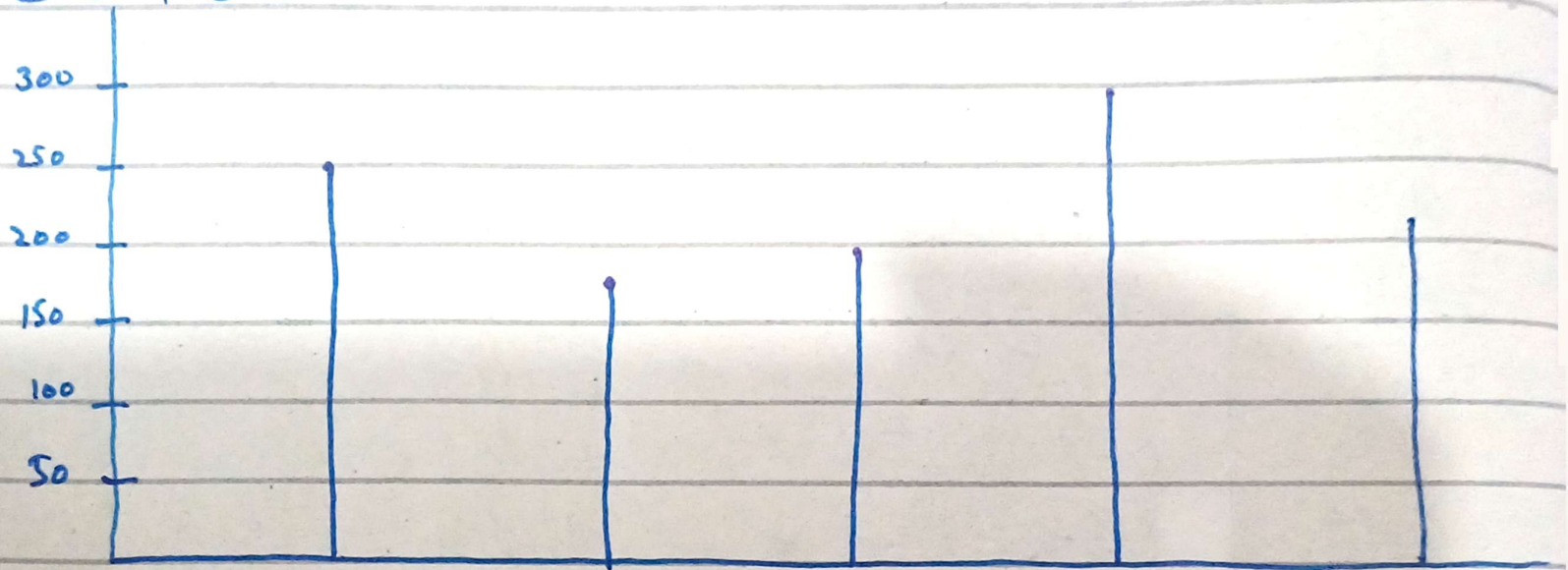
$$\frac{1}{N} \sum (x - \mu)^2 = \frac{1776.84}{12} = 148.07$$

$$\sqrt{\frac{1}{N} \sum (x - \mu)^2} = \sqrt{148.07} = 12.16$$

question 3

Mango	$120 + 130$	$= 250$
Grapes	$85 + 90$	$= 175$
Apples	$95 + 100$	$= 195$
Banana	$140 + 150$	$= 290$
Orange	$110 + 105$	$= 215$

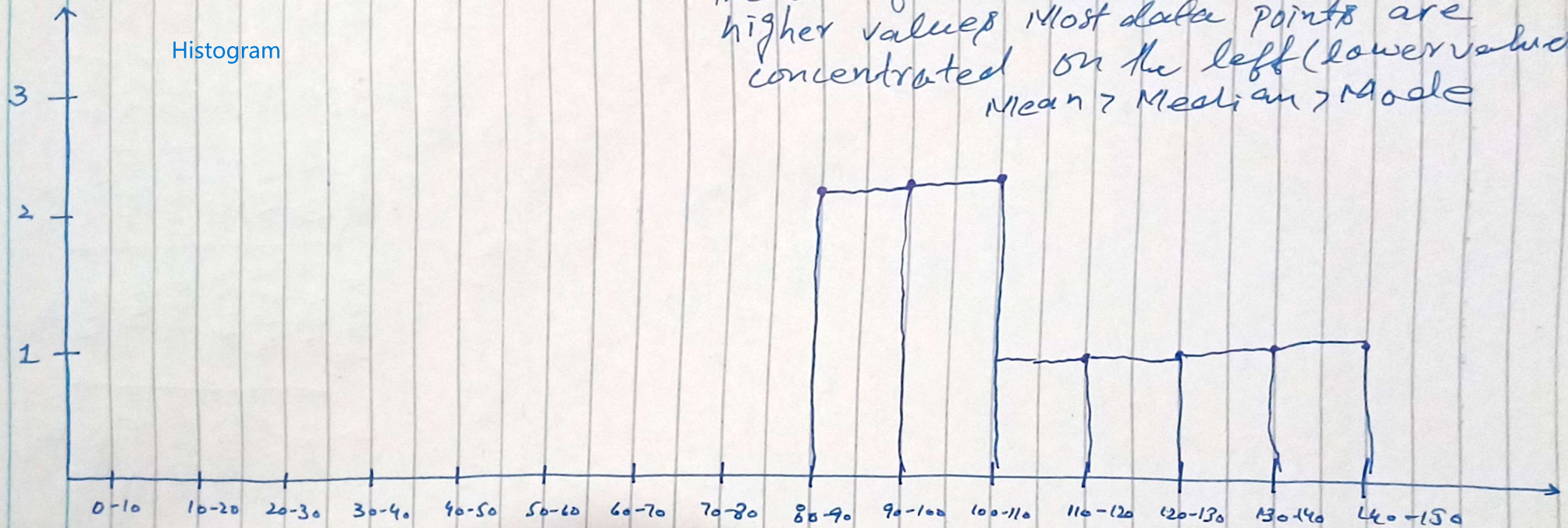
Bar Plot



Mango Grapes Apples Banana Orange

Banana is a top saled fruit

Histogram



The tail of distribution extend towards higher values Most data points are concentrated on the left (lower values)
 $\text{Mean} > \text{Median} > \text{Mode}$

Arrange the data in ascending order

85, 90, 95, 100, 105, 110, 120, 130, 140, 150

$$\text{Mean} = \frac{85 + 90 + 95 + 100 + 105 + 110 + 120 + 130 + 140 + 150}{10} = 112.5$$

$$\text{Median} = \frac{105 + 110}{2} = 107.5$$

$$\text{Mode} = \text{None}$$

$\text{Mean} > \text{Median} > \text{Mode}$

Positively skewed

question 4

X	Y	XY	x^2	y^2
43	99	4257	1849	9801
21	65	1365	441	4225
25	79	1975	625	6241
42	75	3150	1764	5625
57	87	4959	3249	7569
59	81	4779	3481	6561
247	486	20485	11409	40022
$(247)^2$	$(486)^2$			
= 61009	236196			

$$\frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$\frac{6 \times 20485 - (247)(486)}{\sqrt{[6 \times 11409 - 61009][6(40022) - (236196)]}}$$

$$= \frac{2868}{\sqrt{(7445)(3936)}} = \frac{2868}{5413.272}$$

Correlation = 0.5298

* Moderate positive linear relationship

Covariance

X	Y	$X - \bar{X}$	$Y - \bar{Y}$
43	99	$43 - 41.17 = 1.83$	$99 - 81 = 18$
21	65	$21 - 41.17 = -20.17$	$65 - 81 = -16$
25	79	$25 - 41.17 = -16.17$	$79 - 81 = -2$
42	75	$42 - 41.17 = 0.83$	$75 - 81 = -6$
57	87	$57 - 41.17 = 15.83$	$87 - 81 = 6$
59	81	$59 - 41.17 = 17.83$	$81 - 81 = 0$

$$\bar{X} = \frac{43 + 21 + 25 + 42 + 57 + 59}{6} = 41.17$$

$$\bar{Y} = \frac{99 + 65 + 79 + 75 + 87 + 81}{6} = 81$$

$$(X - \bar{X}) * (Y - \bar{Y})$$

1.83	*	18	=	32.94
-20.17	*	-16	=	322.72
-16.17	*	-2	=	32.34
0.83	*	-6	=	-4.98
15.83	*	6	=	94.98
17.83	*	0	=	0

$$\sum (X - \bar{X})(Y - \bar{Y}) = 32.94 + 322.72 + 32.34 + (-4.98) + 94.98 + 0 = 478$$

$$\frac{1}{n} \sum (X - \bar{X})(Y - \bar{Y}) = \frac{478}{6} = 79.67$$

* Covariance sign is positive meaning age increase glucose increase (Directly relationship)

→ No covariance depends on units and scale. High covariance occurs if numbers are large but relationship may not be perfectly linear.